

Tyler Kei

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EDUCATION

Purdue University

B.S. in Computer Science, B.S. in Applied Statistics

Cumulative GPA: 3.42 / 4.00

Relevant Coursework: Data Structures & Algorithms, Large-Scale Data Analytics, Data Mining & Machine Learning, Applied Regression Analysis, Data Science Capstone, Human-Computer Interaction

Aug. 2022 – May 2026

West Lafayette, IN

SKILLS

Languages: Python, Java, JavaScript, SQL, R, HTML/CSS, Bash

Developer Tools: Azure, Databricks, AWS, Git/GitHub, Docker, Power BI, ETL Pipelines, YAML Configuration, LLM Agents

Libraries: React, PyTorch, Pandas, Streamlit, Scikit-Learn, PySpark, XGBoost, NumPy, SHAP, Matplotlib

Certifications: Microsoft Fabric Analytics Engineer (DP-600), Azure Fundamentals (AZ-900), AWS Cloud Practitioner

Interests: Volleyball, Basketball, Skateboarding, FPS, Cooking, Streetwear

EXPERIENCE

Artificial Intelligence Fellow

Eli Lilly – M&Q

May 2026 – Present

Indianapolis, IN

- Built an agent observability layer in Databricks and a custom notification application, monitoring 2 productionized agents, logging 100+ incidents, and supporting 10 developers who maintain them, by calculating agentic evaluation metrics and flagging irregular behavior
- Retired legacy systems by migrating 25 agents and 86 connections into an intelligent, self-formatting graph dashboard that tracks the live status of productionized agents along with their associated data connections
- Built a 3,000 node SharePoint knowledge base using Claude skills and graphify semantic linking, connected via a Microsoft connector to reduce search times and surface deeper insights, then reused those skills to deliver a self-healing FAQ assistant giving 5+ project managers accessibility to current project status

Algorithms Team Lead

Purdue Autonomous Robotics Club

Aug. 2025 – May 2026

West Lafayette, IN

- Led 4 person algorithms team, coordinating weekly integration with mechanical and electrical subteams on servo timing and actuation constraints
- Designed a Viterbi based motion planner converting structured input to motor commands, using lookahead logic to prevent unreachable configurations that failed 15% of passages
- Built a multifactor verification pipeline catching constraint violations before hardware execution
- Implemented split point optimization over 50 candidates, reducing hand movement by 35% versus static assignment

Undergraduate Data Science Researcher

Rolls-Royce

Jan. 2026 – May 2026

West Lafayette, IN

- Refactored a transformer ML pipeline, externalizing 15+ hyperparameters into YAML configs with 6 CLI flags for single command reproducible execution
- Built an auto-checkpointing system persisting model weights, optimizer state, and scheduler every 5 epochs, enabling seamless training resumption after interruptions
- Designed a Streamlit frontend with a custom CSS design system wiring 3 backend ML modules to UI controls, supporting reactive data manipulation across interactive visualizations

PROJECTS

PokeMe

TypeScript, Three.js, React, Zustand, Vite, Tailwind CSS

Aug. 2026 - Present

- Built my personal portfolio website as a playable Pokémon style game spanning 12 connected maps by writing a 3D tile-map engine from scratch in strict mode TypeScript and Three.js with a requestAnimationFrame game loop, instanced meshes, and billboarded sprites
- Kept the Vite production bundle under 250 KB gzipped with zero 3D model or texture files by generating all world art procedurally with the Canvas 2D API at runtime and loading remaining assets in parallel with Promise.all
- Made new areas and portfolio exhibits addable through data alone, with no engine changes, by designing a typed map format (ground grids, props, warps, walk-in triggers) and writing cutscenes as async/await scripts kept separate from a React and Zustand UI